A PROOF OF REDUCE

We prove that the Cluster Partition Problem is NP-hard for every fixed redundancy $m \geq 3$. The proof establishes an equivalence with the rainbow coloring problem on uniform hypergraphs, whose NP-hardness is well known.

Definition A.1 (Cluster Partition Problem). Given a vertex set V, a collection of clusters $\mathcal{C} = \{C_1, \ldots, C_K\}$ where each $C_j \subseteq V$, and a positive integer m such that every vertex $v \in V$ appears in exactly m clusters, the Cluster Partition Problem asks: does there exist a partition of $\mathcal{C}$ into three disjoint subsets S_1, S_2, S_3 such that

$$\bigcup_{C \in S_i} C = V \quad \forall\, i \in \{1, 2, 3\}$$

We denote this problem by $\mathrm{CP}(m)$ when the redundancy is exactly m.

Definition A.2 (Rainbow Coloring). Let $\mathcal{H} = (W, \mathcal{E})$ be a k-uniform hypergraph, i.e., $|e| = k$ for all $e \in \mathcal{E}$. An r-rainbow coloring of $\mathcal{H}$ is a mapping $\chi \colon W \to [r]$ such that for every $e \in \mathcal{E}$,

$$\big\{\chi(w) : w \in e\big\} \supseteq [r],$$

where $[r] = \{1, \ldots, r\}$. The decision problem $\mathrm{RB}(k, r)$ asks whether a given k-uniform hypergraph admits an r-rainbow coloring.

Lemma A.3 (Equivalence via Duality). *$CP(m)$ and $RB(m, 3)$ are polynomial-time equivalent.*

Proof. We show a bijection between instances of the two problems via hypergraph duality.

Direction 1. Given an instance $(V, \mathcal{C})$ of $\mathrm{CP}(m)$, construct a hypergraph $\mathcal{H} = (W, \mathcal{E})$:

- Set $W = \mathcal{C}$ (each cluster becomes a node).
- For each $v \in V$, define $e_v = \{C \in \mathcal{C} : v \in C\}$.

Since v belongs to exactly m clusters, $|e_v| = m$, so $\mathcal{H}$ is m-uniform.

A partition $\mathcal{C} = S_1 \sqcup S_2 \sqcup S_3$ defines a coloring $\chi \colon W \to \{1, 2, 3\}$ via $\chi(C) = i \Leftrightarrow C \in S_i$. The condition "$S_i$ covers V" means that for every v, some $C \in e_v$ receives color i. Requiring this for all three colors is precisely the rainbow condition on e_v.

Direction 2. Given an m-uniform hypergraph $\mathcal{H} = (W, \mathcal{E})$, set $V = \mathcal{E}$ and $\mathcal{C} = W$, where cluster w contains vertex e iff $w \in e$. Since $|e| = m$, each vertex appears in exactly m clusters. By the same reasoning, a 3-rainbow coloring of $\mathcal{H}$ corresponds to a valid partition (S_1, S_2, S_3).

Both directions run in polynomial time. □

Theorem A.4. *For every fixed integer $m \geq 3$, $CP(m)$ is NP-complete.*

Proof. *Membership in NP.* Given (S_1, S_2, S_3), one verifies in polynomial time that each S_i covers V.

NP-hardness. By Lemma A.3, it suffices to show that $\mathrm{RB}(m, 3)$ is NP-hard for every $m \geq 3$.

It is well known that for all $k \geq 3$ and $r \geq 2$, deciding whether a k-uniform hypergraph admits an r-rainbow coloring is NP-hard. This follows from the NP-completeness of Monotone NAE-SAT via Schaefer's dichotomy theorem [48]; see also [15] for foundational hardness results on hypergraph coloring, and [24], where the NP-hardness of $\mathrm{RB}(k, r)$ for all $k \geq 3$, $r \geq 2$ is stated as a known fact.

Setting $r = 3$ and $k = m$, we obtain that $\mathrm{RB}(m, 3)$ is NP-hard for every $m \geq 3$. The result then follows from Lemma A.3. □